

pfSense Frontend Practice – Hands-On Assignment & Assessment Document

LAB_1

```
inet 192.168.143.3 netmask 0xffffffff broadcast 192.168.143.255
inet6 fe80::a00:27ff:fe96:4b06%em1 prefixlen 64 scopeid 0x2
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=23<PERFORMNUD,ACCEPT_RTADV,AUTO_LINKLOCAL>
enc0: flags=0 metric 0 mtu 1536
options=0
groups: enc
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
lo0: flags=1008049<UP,LOOPBACK,RUNNING,MULTICAST,LOWER_UP> metric 0 mtu 16384
options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_IPV6>
inet 127.0.0.1 netmask 0x0
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x4
groups: lo
nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
pflog0: flags=100<PROMISC> metric 0 mtu 33152
options=0
groups: pflog
pfsync0: flags=0 metric 0 mtu 1500
options=0
maxupd: 128 defer: off version: 1400
syncok: 1
groups: psync
[2.8.1-RELEASE][root@pfSense.home.arpal]/root: █
```

The IPv4 LAN address has been set to dhcp

The IPv6 LAN address has been set to dhcp6

Press <ENTER> to continue.

VirtualBox Virtual Machine - Netgate Device ID: 184267e50d3296184996

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

```
WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
                        v6/DHCP6: fd17:625c:f037:2:a00:27ff:fed2:d339/64
LAN (lan) -> em1 -> v4/DHCP4: 192.168.143.3/24
```

- | | |
|-------------------------------------|----------------------------------|
| 0) Logout / Disconnect SSH | 9) pfTop |
| 1) Assign Interfaces | 10) Filter Logs |
| 2) Set interface(s) IP address | 11) Restart GUI |
| 3) Reset admin account and password | 12) PHP shell + pfSense tools |
| 4) Reset to factory defaults | 13) Update from console |
| 5) Reboot system | 14) Enable Secure Shell (sshd) |
| 6) Halt system | 15) Restore recent configuration |
| 7) Ping host | 16) Restart PHP-FPM |
| 8) Shell | |

Enter an option: █

```
Enter the WAN interface name or 'a' for auto-detection
(em0 em1 or a): wannnn
```

```
Invalid interface name 'wannnn'
```

```
Enter the WAN interface name or 'a' for auto-detection
(em0 em1 or a): uplink
```

```
Invalid interface name 'uplink'
```

```
Enter the WAN interface name or 'a' for auto-detection
(em0 em1 or a): em0
```

```
Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(em1 a or nothing if finished): em1
```

```
The interfaces will be assigned as follows:
```

```
WAN -> em0
LAN -> em1
```

```
Do you want to proceed [y/n]? y
```

```

    inet 192.168.143.3 netmask 0xffffffff broadcast 192.168.143.255
    inet6 fe80::a00:27ff:fe96:4b06%em1 prefixlen 64 scopeid 0x2
    media: Ethernet autoselect (1000baseT <full-duplex>)
    status: active
    nd6 options=23<PERFORMNUD,ACCEPT_RTADV,AUTO_LINKLOCAL>
enc0: flags=0 metric 0 mtu 1536
    options=0
    groups: enc
    nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
lo0: flags=1008049<UP,LOOPBACK,RUNNING,MULTICAST,LOWER_UP> metric 0 mtu 16384
    options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPV6,TXCSUM_IPV6>
    inet 127.0.0.1 netmask 0x0
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x4
    groups: lo
    nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
pflog0: flags=100<PROMISC> metric 0 mtu 33152
    options=0
    groups: pflog
pfsync0: flags=0 metric 0 mtu 1500
    options=0
    maxupd: 128 defer: off version: 1400
    syncok: 1
    groups: psync
[2.8.1-RELEASE][root@pfSense.home.arpal]/root: █
```

```

    groups: pflog
pfsync0: flags=0 metric 0 mtu 1500
    options=0
    maxupd: 128 defer: off version: 1400
    syncok: 1
    groups: pfsync
[2.8.1-RELEASE][root@pfSense.home.arpal/root: if config em0
if: Expression Syntax.
[2.8.1-RELEASE][root@pfSense.home.arpal/root: if config em0 down
if: Expression Syntax.
[2.8.1-RELEASE][root@pfSense.home.arpal/root: ifconfig em0 down
[2.8.1-RELEASE][root@pfSense.home.arpal/root: ifconfig em0
em0: flags=1008c02<BROADCAST,DRV_OACTIVE,SIMPLEX,MULTICAST,LOWER_UP> metric 0 mt
u 1500
    options=48100bb<RXCSUM, TXCSUM, VLAN_MTU, VLAN_HWTAGGING, JUMBO_MTU, VLAN_HWC
SUM, VLAN_HWFILTER, HWSTATS, MEXTPG>
    ether 08:00:27:d2:d3:39
    inet 10.0.2.15 netmask 0xffffffff00 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fed2:d339%em0 prefixlen 64 tentative scopeid 0x1
    inet6 fd17:625c:f037:2:a00:27ff:fed2:d339 prefixlen 64 tentative autocon
f pltime 14400 vlttime 86400
    media: Ethernet autoselect (1000baseT <full-duplex>)
    status: active
    nd6 options=23<PERFORMNUD,ACCEPT_RTADV,AUTO_LINKLOCAL>
[2.8.1-RELEASE][root@pfSense.home.arpal/root:

```

LAB_2

pfSense.home.arpa - Firewall: Rules

Versions of pfSense software available

Interface Types and Configurations

New Tab

192.168.143.3/firewall_rules.php?if=lan

Firewall / Rules / LAN

The firewall rule configuration has been changed.
The changes must be applied for them to take effect.

Apply Changes

Floating WAN LAN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	6/126 KIB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	
☐	0/0 B	IPv4 *	192.168.10.50	*	*	*	*	none			
☐	0/0 B	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
☐	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

Add Add Delete Toggle Copy Save Separator

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18°C Sunny

ENG IN 10:14 AM 13-12-2025

pfSense.home.arpa - Firewall Rules

Versions of pfSense software available

Interface Types and Configurations

New Tab

Not secure 192.168.143.3/firewall_rules.php?if=lan

The firewall rule configuration has been changed.
The changes must be applied for them to take effect.

Apply Changes

Floating

WAN

LAN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	5/165 KiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	
☐	0/0 B	IPv4 ICMP any	*	*	*	*	*	none		block icmp from lan	
☐	0/0 B	IPv4 *	192.168.10.50	*	*	*	*	none			
☐	0/0 B	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
☐	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

↑ Add

↓ Add

Delete

Toggle

Copy

Save

+ Separator

i

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pfSense.home.arpa - Firewall Schedules

Not secure 192.168.143.3/firewall_schedule.php

pfSense COMMUNITY EDITION

System Interfaces Firewall Services VPN Status Diagnostics Help

WARNING:
The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.

Firewall / Schedules

?

Schedules

Name	Range: Date / Times / Name	Description	Actions
Working	Mon - Sat / 9:00-17:59 /	. Create a time-based rule allowing traffic only during working hours.	

Indicates that the schedule is currently active.

+ Add

i

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LAB_3

The screenshot displays the pfSense Firewall NAT Outbound configuration page. The top section, 'Outbound NAT Mode', shows four radio button options: 'Automatic outbound NAT rule generation. (IPsec passthrough included)' (selected), 'Hybrid Outbound NAT rule generation. (Automatic Outbound NAT + rules below)', 'Manual Outbound NAT rule generation. (AON - Advanced Outbound NAT)', and 'Disable Outbound NAT rule generation. (No Outbound NAT rules)'. A 'Save' button is located below these options.

The 'Mappings' section contains a table with columns: Interface, Source, Source Port, Destination, Destination Port, NAT Address, NAT Port, Static Port, Description, and Actions. The table is currently empty, with 'Add', 'Add', 'Delete', 'Toggle', and 'Save' buttons at the bottom right.

The 'Automatic Rules' section shows a table with columns: Interface, Source, Source Port, Destination, Destination Port, NAT Address, NAT Port, Static Port, and Description. It lists two rules for the WAN interface, both with source 127.0.0.0/8 and destination 192.168.143.0/24. The first rule has destination port 500 and is described as 'Auto created rule for ISAKMP'. The second rule has an asterisk for destination port and is described as 'Auto created rule'.

Below the configuration page, a notification banner states: 'The NAT configuration has been changed. The changes must be applied for them to take effect.' with an 'Apply Changes' button.

The bottom section shows the 'Port Forward' tab selected, with sub-tabs for '1:1', 'Outbound', and 'NPT'. The 'Outbound NAT Mode' section is repeated, showing the same four radio button options, with 'Automatic outbound NAT rule generation. (IPsec passthrough included)' still selected. A 'Save' button is present.

The 'Mappings' section now contains one entry: WAN interface, source 192.168.143.0/24, source port asterisk, destination asterisk, destination port asterisk, NAT address WAN address, NAT port asterisk, static port icon, and description 'Configure outbound NAT for one internal subnet only'. 'Add', 'Add', 'Delete', 'Toggle', and 'Save' buttons are at the bottom right.

The 'Automatic Rules' section is also repeated, showing the same two rules for the WAN interface.